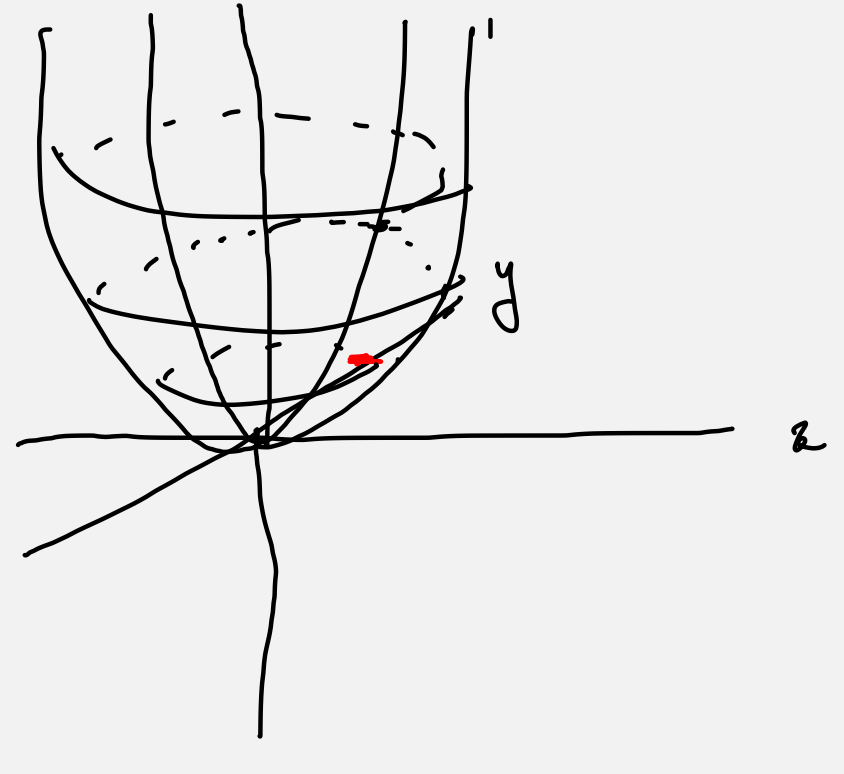
$f: U \subset \mathbb{R}^n$, $C_a = \{p \in U : f(p) = a\}$

$\underline{1-}$ $f: (x,y) \longrightarrow x^2+y^2$.



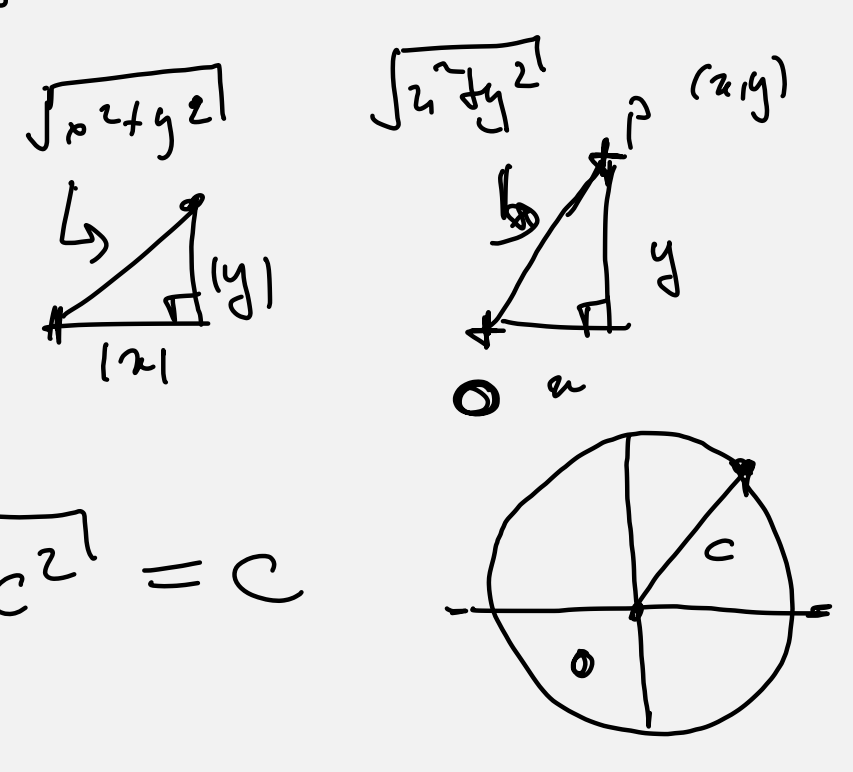
$\{y=0\}$ $f_x: x \longrightarrow x^2+0$
$\{x=0\}$ $f_y: y \longrightarrow y^2$
$\{y=c\}$ $f_c: x \longrightarrow x^2+c^2$.

$\mathbb{R}^2$ = dominio

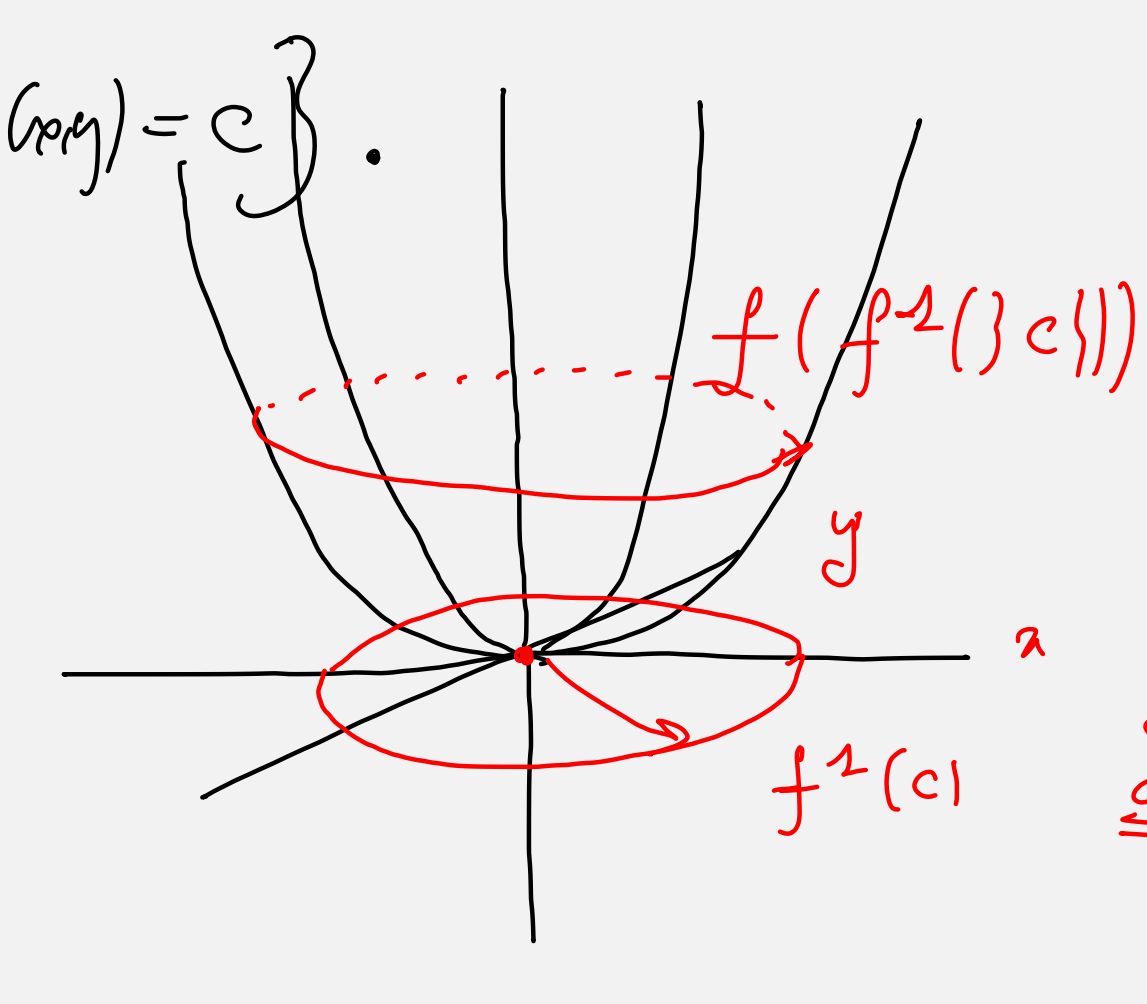
$f^{-1}(\{c\}) = \{(x,y) \in \mathbb{R}^2 \mid f(x,y) = c\}$.

(x,y) tq $x^2+y^2 = c^2$

$\|\cdot\|_2^2$



$c>0$ $\sqrt{x^2+y^2} = \sqrt{c^2} = c$



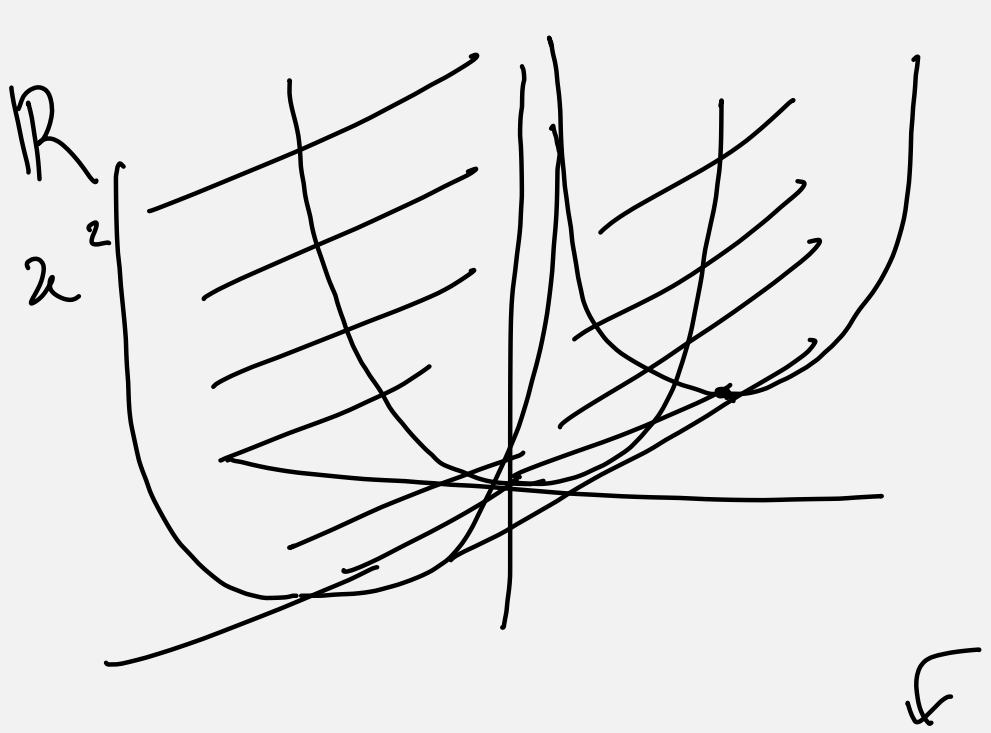
$f(f^{-1}(\{c\})) = \{f(p) \mid p \in f^{-1}(c)\}$.

$\underline{\underline{Si}}$ $c \geq 0$

$f(p) = 0 \iff x^2+y^2 = 0 \Rightarrow x^2 = y^2 = 0 \Rightarrow \underline{x=y=0}$.

$c<0$ no van a existir $(x,y) \in \mathbb{R}^2$ tq $f(x,y) < 0$

$f: \mathbb{R}^2 \longrightarrow \mathbb{R}$
$(x,y) \longrightarrow x^2$

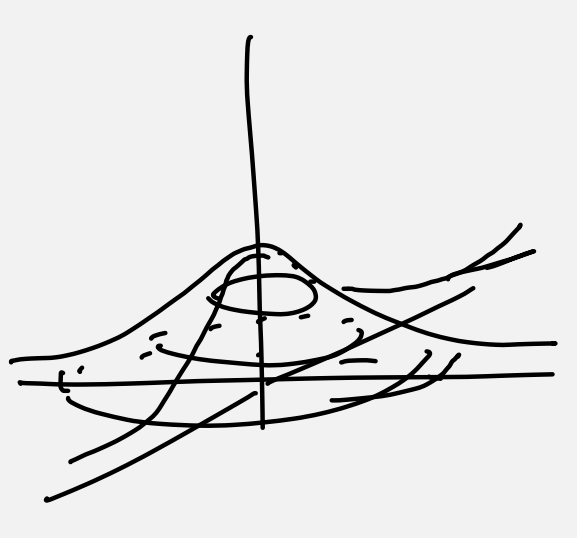


$r^2 = d(0,(x,y))^2$

$\int_{\mathbb{R}} e^{-x^2} dx = \underset{\smile}{c} \cdot \sqrt{\pi}$

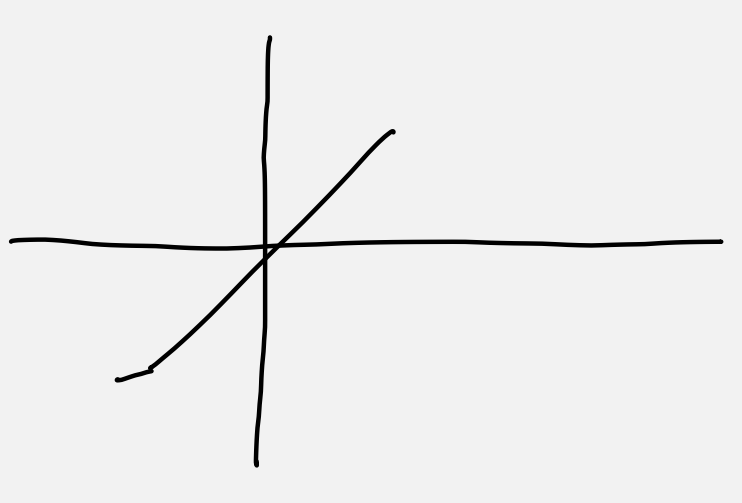
$\mathbb{N}$
$\mathbb{Z}$
$\mathbb{Q}$

$\int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx\, dy$



$\underline{2-a-}$ $f: (x,y,z) \longrightarrow \frac{x}{x-y-z}$

$\mathbb{R}^3 \longrightarrow \mathbb{R}$



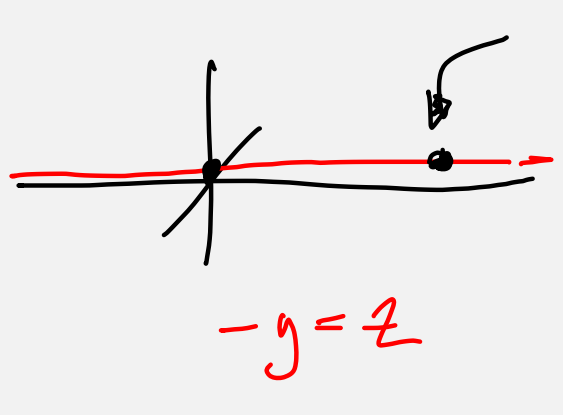
$\{x-y-z \neq 0\} = D_f$.

$Adh(D_f)$. f C^0 es: $x_n \longrightarrow x \Rightarrow f(x_n) \longrightarrow f(x)$,

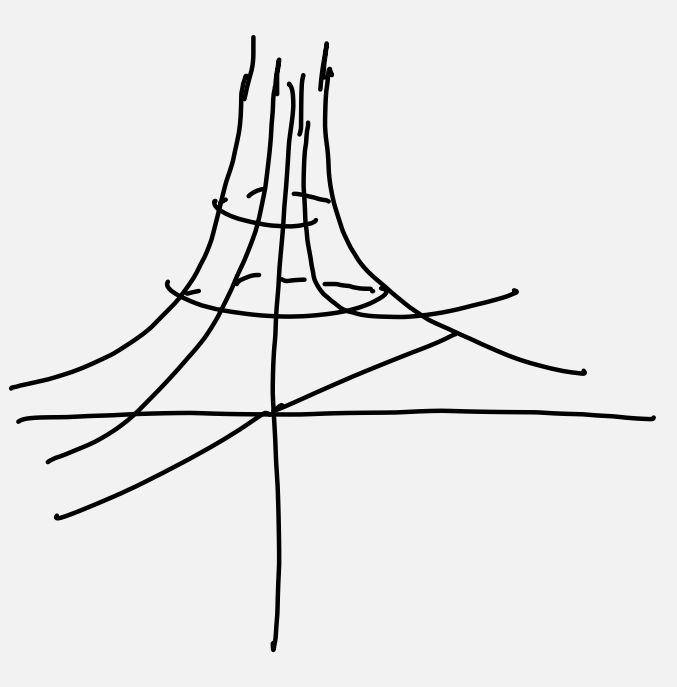
$\{q \text{ tq } \exists p_n \xrightarrow{n\to\infty} q\}$ $\tilde{f} = \begin{cases} f & \text{en } D_f \\ \lim f(p_n) & \text{cuando } p_n \longrightarrow p \in Adh(D_f) \end{cases}$

$\forall (p_n) \in \mathbb{R}^n$ tq $p_n \longrightarrow p$, si $\lim f(p_n)\ \exists$

$f_{x=0}(x,y,z) = 0$

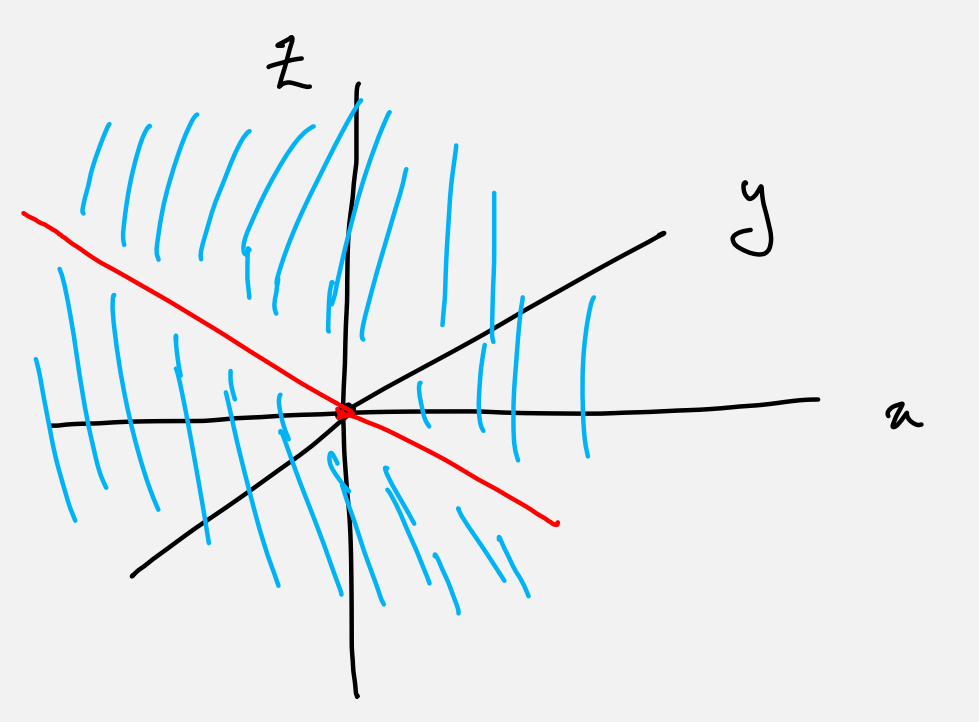


$f: (x,y) \longrightarrow \frac{1}{x^2+y^2}$



$\underline{f(x,y,z) = c:}$ $f(x,y,z) = 0 \iff \frac{x}{\underbrace{x-y-z}_{\neq 0}} = 0 \iff x = 0$

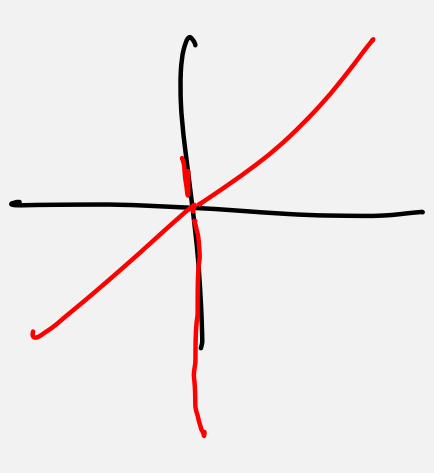
$\begin{cases} x=0 \\ x-y-z \neq 0 \end{cases} \iff \begin{cases} x=0 \\ y \neq -z \end{cases}$



$\underline{c \neq 0}$: $\frac{x}{x-y-z} = c \iff x = c(x-y-z)$ (C)

$x-y-z \neq 0$ (c) $\iff 0 = (c-1)x - cy - cz$

$L\ \mathbb{R}^2$: $ax+by=0$
$(b \neq 0)$: $y = -\frac{a}{b}x$



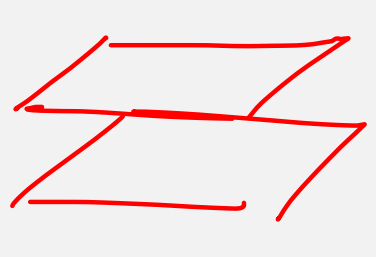
- $x-y$

$z=0$?

$x=y$

$y=x-1$

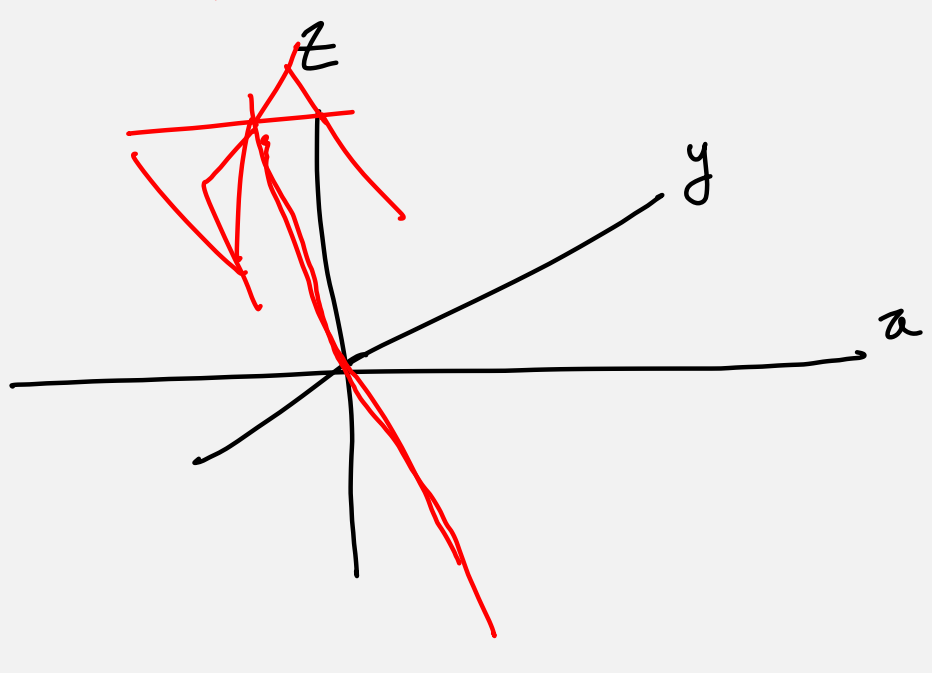
$\begin{cases} 0=(c-1)x-cy-cz \\ x \neq y+z \end{cases}$ $x=y+z$



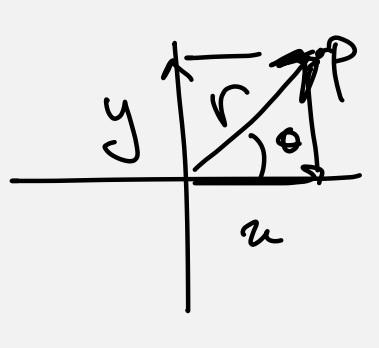
③. (x,y,z)
$\begin{cases} 0 = (c-1)\cdot x - cy - cz & ① \\ 0 = +x-y-z & ② \end{cases}$

① − c ② : $0 = (c-1)x + cx$
$(2c-1)x = 0 \Rightarrow x=0$

$\begin{cases} 0 = c(y+z) \\ 0 = y+z \end{cases} \Rightarrow y+z=0$

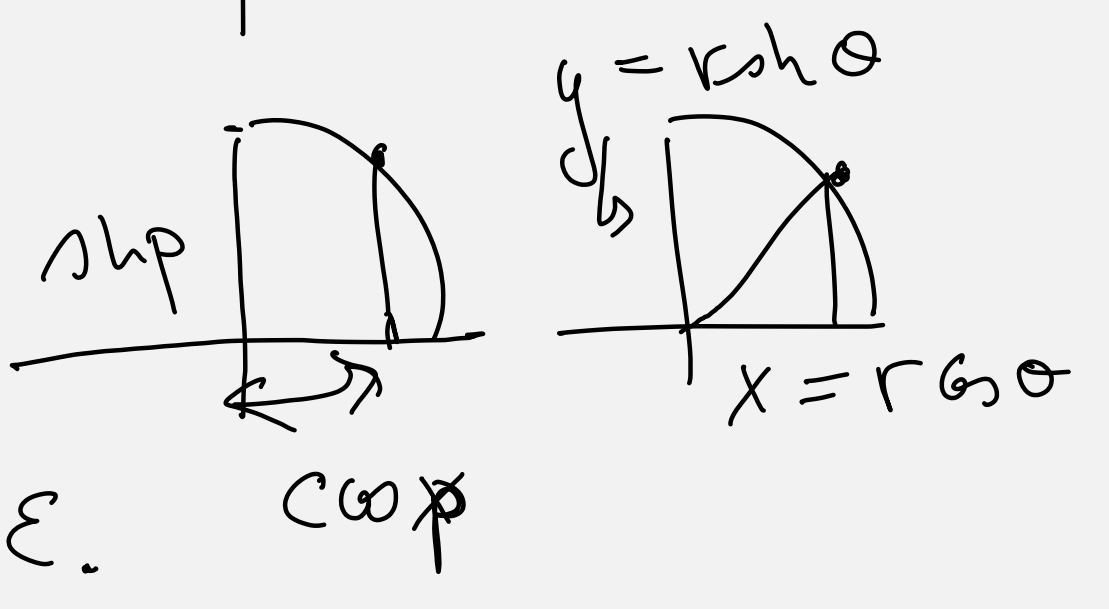


$\underline{4-}$ $f: U \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}$. $X = r\cos\theta$
f C^0. $U = B_R((0,0))$ $y = r\sin\theta$



$\theta \in [0, 2\pi[$

$\underline{a-}$ Pg $\lim_{(x,y)\to(0,0)} f(x,y) = L \iff$



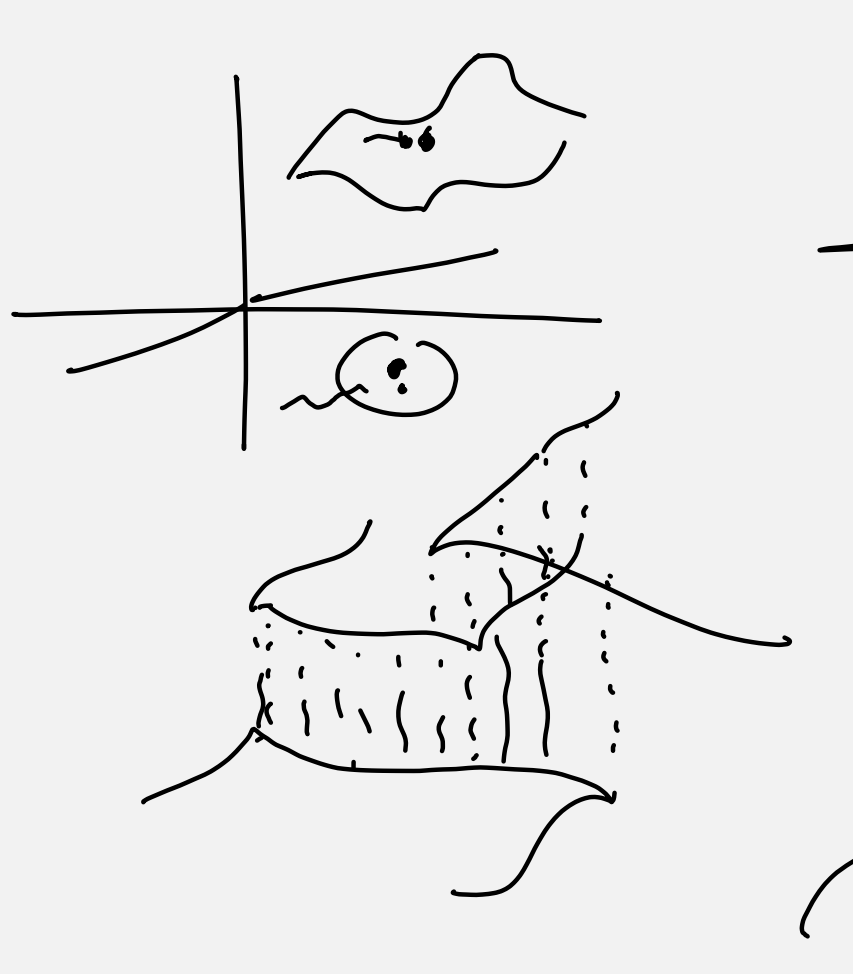
$\iff \forall \varepsilon>0, \exists \delta>0$ tq $|f(r\cos\theta, r\sin\theta) - L| \leq \varepsilon$.

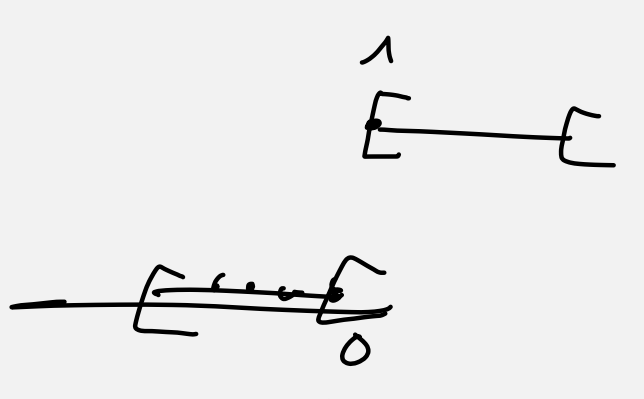
• $\lim \|(x,y)-(0,0)\|_2 = 0$

$(r,\theta) \longrightarrow (r\cos\theta, r\sin\theta)$ es C^0
$\mathbb{R}_+ \times [0,2\pi[\longrightarrow \mathbb{R}^2$

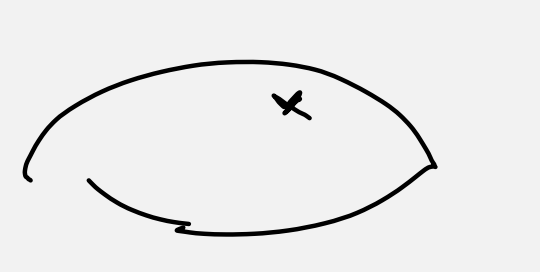
f es C^0.

f C^0 en (a,b): $\forall \varepsilon>0, \exists \delta>0$ tq $\forall (x,y) \in D_f$, $\|(a,b)-(x,y)\|_2 \leq \delta$
$\Rightarrow |f(a,b) - f(x,y)| \leq \varepsilon$





$\lim f(x_n) \neq f(x)$



① ②

$\lim_{(x,y)\to(0,0)} f(x,y) = L \iff \dots$

① $:= \forall \varepsilon>0, \exists \eta>0$ tq $\|(x,y)-(0,0)\|_2 \leq \eta \Rightarrow |f(x,y)-(0,0)|$ [illegible] ε

$(r,\theta) \longrightarrow r\cos(\theta)$
$\mathbb{R}_+ \times [0,2\pi[$ $\mathbb{R}$

f_1, f_2. $(x,y) \longrightarrow (f_1(x,y), f_2(x,y))$
$\|\cdot\|_2$

$(r,\theta) \longrightarrow \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix} \longrightarrow f(r\cos\theta, r\sin\theta)$

$\forall \eta>0, \exists \delta>0$ tq $|0-r| \leq \delta \overset{①}{\Rightarrow} \underbrace{\|(r\cos\theta, r\sin\theta)-(0,0)\|_2 \leq \eta}$

$\Rightarrow |f(r\cos\theta, r\sin\theta) - \underbrace{f(0,0)}_{=L}| \leq \varepsilon$

① $\sqrt{r^2\cos^2\theta + r^2\sin^2\theta - 0} = \sqrt{r^2(\underbrace{\cos^2+\sin^2}_{=1})}$
$= r$